ELEVTRBL - Elevator Trouble

#bfs (/problems/tag/bfs)

You are on your way to your first job interview as a program tester, and you are already late. The interview is in a skyscraper and you are currently in floor *s*, where you see an elevator. Upon entering the elevator, you learn that it has only two buttons, marked "UP *u*" and "DOWN *d*". You conclude that the UP-button takes the elevator *u* floors up (if there aren't enough floors, pressing the UP-button does nothing, or at least so you assume), whereas the DOWN-button takes you *d* stories down (or none if there aren't enough). Knowing that the interview is at floor *g*, and that there are only *f* floors in the building, you quickly decide to write a program that gives you the amount of button pushes you need to perform. If you simply cannot reach the correct floor, your program halts with the message "`use the stairs`".

Given input *f*, *s*, *g*, *u* and *d* (floors, start, goal, up, down), find the shortest sequence of button presses you must press in order to get from *s* to *g*, given a building of floors, or output "`use the stairs`" if you cannot get from *s* to *g* by the given elevator.

Input

The input will consist of one line, namely `f s g u d`, where 1 ≤ s, g ≤ f ≤ 1000000 and 0 ≤ u, d ≤ 1000000. The floors are one-indexed, i.e. if there are 10 stories, s and g be in [1; 10].

Output

You must reply with the minimum numbers of pushes you must make in order to get from s to g, or output "`use the stairs`" if it is impossible given the conguration of the elevator.

Example

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Input:
10 1 10 2 1

Output:
6
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